

OPERA NUMERORUM

NS Tower -- Path A: Phase 105
NS_BlowupConcentration_PROVED -- ESS Blowup Profile

Author: David Fox | ORCID: 0009-0008-1290-6105 | Date: July 2, 2026

THEOREM

IF u blows up at T^* , then there exists a nonzero ancient NS solution u_{inf} in $L^{\{3,\text{inf}\}}$. Proof route: parabolic rescaling (Phase 103) + Banach-Alaoglu + Caffarelli-Kohn-Nirenberg concentration.

MATHEMATICAL PROOF (KEY STEPS)

Step 1 (Scale invariance): $u_k(x,t) = \lambda_k u(\lambda_k x, T^* + \lambda_k^2 t)$ solves NS (Phase 103). Step 2 (Uniform bound): $\|u_k\|_{L^{\{3,\text{inf}\}}} = \|u\|_{L^{\{3,\text{inf}\}}}$ (scale-invariant norm). Step 3 (Compactness): Banach-Alaoglu: bounded sequence in $L^{\{3,\text{inf}\}}(K)$ has weak-* convergent subsequence. Step 4 (Concentration): Caffarelli-Kohn-Nirenberg: blowup time forces $\|u_{\text{inf}}\|_{L^{\{3,\text{inf}\}}(B_1 \times (-1,0))} \geq \epsilon_0 > 0$. Step 5 (Ancient solution): u_{inf} defined on all t in $(-\text{inf}, 0]$, solves NS (closure under weak-* limits). CONCLUSION: Nonzero ancient NS solution u_{inf} exists IF blowup occurs. Contradiction will come in Phase 108.

DEP COUNT: 4 -> 3 deps remaining after this phase

#	Remaining Dep	Status
1	NS_Carleman_LimitPass_OPEN	OPEN (after Phases 106-107)
2	NS_CarlemanHeat_OPEN	OPEN CRITICAL -- Hormander (NEXT Phase 106)
3	NS_CarlemanDriftAbsorption_OPEN	OPEN (after Phase 106)

INTEGRITY

Phase 105 | 0 sorry | 0 axiom keyword | classical trio Cumulative:
95(7)->101(7)->102(6)->103(5)->104(4)->105(3) #print axioms NS_M6_CLOSED_v105 -> {propext,
Classical.choice, Quot.sound}